

Python



University of Papua New Guinea

School of Natural and Physical Sciences

Department of Mathematics, Statistics and Computer Science

Address: PO Box 320 University PO Waigani, Port Moresby, Phone: 326 7414, Email: secmaths@upng.ac.pg

Python tuples:

- A tuple is a list that cannot change.
- Python refers to a value that cannot change as immutable.
- So by definition, a tuple is an immutable list.

Defining a tuple:

A tuple is like a list except that it uses parentheses () instead of square brackets [].

Example:

```
rgb = ('red', 'green', 'blue')
```

```
print(rgb[0])
```

```
print(rgb[1])
```

```
print(rgb[2])
```

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Since a tuple is immutable, you cannot change its elements.

```
rgb = ('red', 'green', 'blue')  
rgb[0] = 'yellow'
```

TypeError: 'tuple' object does not support item assignment

An empty tuple can be created as follows.

```
T4 = ()
```

```
tuple1 = (10, 20, 30, 40, 50, 60)  
print(tuple1)  
count = 0  
for i in tuple1:  
    print("tuple1[%d] = %d"%(count, i))  
    count = count+1
```

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```
tuple1 = tuple(input("Enter the tuple elements ..."))
print(tuple1)
count = 0
for i in tuple1:
    print("tuple1[%d] = %s"%(count, i))
    count = count+1
```

Output:

```
Enter the tuple elements ...123456
('1', '2', '3', '4', '5', '6')
tuple1[0] = 1
tuple1[1] = 2
tuple1[2] = 3
tuple1[3] = 4
tuple1[4] = 5
tuple1[5] = 6
```



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Tuple indexing and slicing:

Tuple = (0, 1, 2, 3, 4, 5)

0	1	2	3	4	5
Tuple[0] = 0	Tuple[0:] = (0, 1, 2, 3, 4, 5)				
Tuple[1] = 1	Tuple[:] = (0, 1, 2, 3, 4, 5)				
Tuple[2] = 2	Tuple[2:4] = (2, 3)				
Tuple[3] = 3	Tuple[1:3] = (1, 2)				
Tuple[4] = 4	Tuple[:4] = (0, 1, 2, 3)				
Tuple[5] = 5					

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```
tuple = (1,2,3,4,5,6,7)
#element 1 to end
print(tuple[1:])
#element 0 to 3 element
print(tuple[:4])
#element 1 to 4 element
print(tuple[1:5])
# element 0 to 6 and take step of 2
print(tuple[0:6:2])
```

```
-----
tuple1 = (1, 2, 3, 4, 5)
print(tuple1[-1])
print(tuple1[-4])
print(tuple1[-3:-1])
print(tuple1[: -1])
print(tuple1[-2:])
```



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Deleting Tuple:

- Unlike lists, the tuple items cannot be deleted by using the del keyword as tuples are immutable.
- To delete an entire tuple, we can use the del keyword with the tuple name.

```
tuple1 = (1, 2, 3, 4, 5, 6)
```

```
print(tuple1)
```

```
del tuple1[0]
```

```
print(tuple1)
```

```
del tuple1
```

```
print(tuple1)
```

```
(1, 2, 3, 4, 5, 6)
```

Traceback (most recent call last):

File "tuple.py", line 4, in <module>

```
print(tuple1)
```

NameError: name 'tuple1' is not defined



List vs. Tuple

SN	List	Tuple
1	The literal syntax of list is shown by the [].	The literal syntax of the tuple is shown by the ().
2	The List is mutable.	The tuple is immutable.
3	The List has the a variable length.	The tuple has the fixed length.
4	The list provides more functionality than a tuple.	The tuple provides less functionality than the list.
5	The list is used in the scenario in which we need to store the simple collections with no constraints where the value of the items can be changed.	The tuple is used in the cases where we need to store the read-only collections i.e., the value of the items cannot be changed. It can be used as the key inside the dictionary.
6	The lists are less memory efficient than a tuple.	The tuples are more memory efficient because of its immutability.

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Python Sort List:

- To sort a list, you use the `sort()` method:

```
list.sort()
```

```
guests = ['James', 'Mary', 'John', 'Patricia', 'Robert', 'Jennifer']  
guests.sort()  
print(guests)
```

- `reverse=True` argument to sort the elements in the `guests` list in the reverse alphabetical order:

```
guests = ['James', 'Mary', 'John', 'Patricia', 'Robert', 'Jennifer']  
guests.sort(reverse=True)  
  
print(guests)
```

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Python List `sort()` method to sort a list of numbers:

```
scores = [5, 7, 4, 6, 9, 8]
scores.sort()
print(scores)
```

```
scores = [5, 7, 4, 6, 9, 8]
scores.sort(reverse=True)
print(scores)
```



Python Set:

A Python set is an unordered list of immutable elements. It means:

- Elements in a set are unordered.
- Elements in a set are unique. A set doesn't allow duplicate elements.
- Elements in a set cannot be changed. For example, they can be [numbers](#), [strings](#), and [tuples](#), but cannot be [lists](#) or [dictionaries](#).

To define a set in Python, you use the curly brace {}. For example:

```
skills = {'Python programming', 'Databases', 'Software design'}
```

To define an empty set, you cannot use the curly braces like this:

```
empty_set = set()
```

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Getting sizes of a set:

```
ratings = {1, 2, 3, 4, 5}
size = len(ratings)
print(size)
```

you can pass a list, which is an iterable, to the set() function like this:

```
skills = set(['Problem solving', 'Critical Thinking'])
print(skills)
```

If an iterable has duplicate elements, the set() function will remove them.
For example:

```
characters = set('letter')
print(characters)
```

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Checking if an element is in a set:

- To check if a set contains an element, you use the in operator:

```
ratings = {1, 2, 3, 4, 5}
rating = 1
```

```
if rating in ratings:
    print(f'The set contains {rating}')
```

To negate the in operator, you use the not operator.
For example:

```
ratings = {1, 2, 3, 4, 5}
rating = 6
```

```
if rating not in ratings:
    print(f'The set does not contain {rating}')
```

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Adding elements to a set

- To add an element to a set, you use the `add()` method:
`set.add(element)`

For example:

```
skills = {'Python programming', 'Software design'}  
skills.add('Problem solving')  
print(skills)
```

Removing an element from a set

To remove an element from a set, you use the `remove()` method:
`set.remove(element)`

For example:

```
skills = {'Problem solving', 'Software design', 'Python programming'}  
skills.remove('Software design')  
print(skills)
```

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Removing all elements from a set

- To remove all elements from a set, you use the `clear()` method:

```
set.clear()
```

```
skills = {'Problem solving', 'Software design', 'Python programming'}  
skills.clear()  
print(skills)
```

Output:

```
set()
```

Frozen a set:

- To make a set immutable, you use the built-in function called `frozenset()`. The **`frozenset()`** returns a new immutable set from an existing one.

For example:

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After that, if you attempt to modify elements of the set, you'll get an error:

```
skills.add('Django')
```

Looping through set elements:

```
skills = {'Problem solving', 'Software design', 'Python programming'}  
for skill in skills:  
    print(skill)
```

Output:

```
Software design  
Python programming  
Problem solving
```

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Python Set Union:

- The union of two sets returns a new set that contains distinct elements from both sets.

Example:

```
s1 = {'Python', 'Java'}
```

```
s2 = {'C#', 'Java'}
```

Output:

```
{ 'Java', 'Python', 'C#' }
```

#Union sets using union() method:

```
s1 = {'Python', 'Java'}
```

```
s2 = {'C#', 'Java'}
```

```
s = s1.union(s2)
```

```
print(s)
```

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Union sets using the | operator:

- Python provides you with the set union operator | that allows you to union two sets:

```
s1 = {'Python', 'Java'}  
s2 = {'C#', 'Java'}  
s = s1 | s2  
print(s)
```

Output:

```
{'Java', 'C#', 'Python'}
```

The union() method vs. set union operator:

- the union() method accepts one or more iterables, converts the iterables to sets, and performs the union.

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```
rates = {1, 2, 3}
ranks = [2, 3, 4]
ratings = rates.union(ranks)
print(ratings)
```

Output:

```
{1, 2, 3, 4}
```

```
rates = {1, 2, 3}
ranks = [2, 3, 4]
ratings = rates | ranks
```

Error:

`TypeError: unsupported operand type(s) for |: 'set' and 'list'`

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Python Set Intersection:

- In Python, you can use the Set intersection() method or set intersection operator (&) to intersect two or more sets:

```
new_set = set1.intersection(set2, set3)
new_set = set1 & set2 & set3
```

```
s1 = {'Python', 'Java', 'C++'}
s2 = {'C#', 'Java', 'C++'}
```

```
s = s1.intersection(s2) // or s = s1 & s2
```

```
print(s)
```

Output:

```
{'C++', 'Java'}
```

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Python Set Difference:

- The difference between the two sets results in a new set that has elements from the first set which aren't present in the second set.

```
s1 = {'Python', 'Java', 'C++'}  
s2 = {'C#', 'Java', 'C++'}  
s = s1.difference(s2)  
print(s)
```

Output:

```
{'Python'}
```

```
s1 = {'Python', 'Java', 'C++'}  
s2 = {'C#', 'Java', 'C++'}  
s = s2.difference(s1)
```

```
print(s)
```

Output:

```
{'C#'}
```

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Python Symmetric Difference:

- The symmetric difference of two sets is a set of elements that are in either set, but not in their intersection.

Example:

```
s1 = {'Python', 'Java', 'C++'}
```

```
s2 = {'C#', 'Java', 'C++'}
```

```
s = s1.symmetric_difference(s2)
```

```
print(s)
```

Output:

```
{'C#', 'Python'}
```

```
s1 = {'Python', 'Java', 'C++'}
```

```
s2 = {'C#', 'Java', 'C++'}
```

```
s = s1 ^ s2
```

```
print(s)
```